

Reach Dog — Complete System Requirements Specification and System Design

MVP Foundation and Scalable Evolution (v2.0)

Reach Dog Project Team

April 2026

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1. Introduction

1.1 Purpose

This document is the complete System Requirements Specification (SRS) and system design for **Reach Dog**, an iOS-native CRM application built for sales professionals who capture leads at in-person events. It covers both the MVP and the scalable production version. It serves as the single source of truth for developers, testers, and stakeholders.

1.2 Scope

Reach Dog is a native iOS application that reduces the time from “business card in hand” to “follow-up email drafted” from hours to seconds. It combines on-device OCR, LLM-powered email generation (Gemini Flash API), and a local-first lead management workspace.

In scope: iOS-native app, Apple/Google sign-in, on-device OCR, AI follow-up generation, lead management, email dispatch via URL schemes, CSV import/export, dashboard analytics.

Out of scope: Android, web app, desktop app, full email client, SMTP sending, bidirectional CRM integration (Salesforce, HubSpot), company-operated backend (MVP).

1.3 Definitions, Acronyms, Abbreviations

Term	Definition
CRM	Customer Relationship Management
MVP	Minimum Viable Product
OCR	Optical Character Recognition
LLM	Large Language Model
OAuth 2.0	Open Authorization protocol, RFC 6749
JWT	JSON Web Token, RFC 7519
SwiftData	Apple’s declarative persistence framework
Vision	Apple’s on-device computer-vision framework
Keychain	Apple’s secure credential-storage service
CloudKit	Apple’s cloud-persistence framework
MVVM	Model-View-ViewModel architectural pattern
DFD	Data-Flow Diagram
ERD	Entity-Relationship Diagram
TLS	Transport Layer Security
HIG	Apple Human Interface Guidelines
URL Scheme	Inter-application communication mechanism on iOS
Lead	A prospective customer captured within the application
ATS	App Transport Security
AES	Advanced Encryption Standard

1.4 References

- IEEE Std 830-1998: Recommended Practice for Software Requirements Specifications
- RFC 6749: The OAuth 2.0 Authorization Framework
- RFC 7519: JSON Web Token (JWT)

- Apple Inc., AuthenticationServices Framework Reference, 2025
- Apple Inc., Vision Framework Reference, 2025
- Apple Inc., SwiftData Programming Guide, 2025
- Apple Inc., CloudKit Framework Reference, 2025
- Apple Inc., Human Interface Guidelines — iOS, 2025
- Google LLC, Gemini API Documentation (Flash Tier), 2026
- Google LLC, GoogleSignIn-iOS SDK Reference, 2026

1.5 Document Overview

Section	Content
1	Introduction
2	Literature Review
3	SWOT and PESTLE Analysis
4	Planning Phase
5	Methodology
6	Functional and Non-Functional Requirements
7	Process and Data Modelling
8	System Design
9	UML Diagrams (all 11 types)
10	MVP Design
11	Testing
12	Deployment Plan
13	Maintenance Plan
14	References

2. Literature Review

2.1 Existing Solutions

Tool	Strengths	Weaknesses (vs Reach Dog's use case)
Salesforce Mobile	Full CRM, enterprise features	Overkill for individual reps; no OCR; high cost
HubSpot CRM	Free tier, web-based	No native OCR; requires web connectivity; not iOS-native
CamCard	Business card OCR	No AI follow-up; no lead pipeline; dated UI
Covve	Contact management + card scan	No AI email drafting; limited pipeline
Apple Contacts	Built-in, free	No CRM pipeline; no AI; no follow-up workflow

2.2 Gap Identified

No existing tool combines **on-device OCR + AI-generated personalised follow-up emails + lead pipeline management** in a single native iOS app with local-first data ownership. Reach Dog fills this gap by targeting the specific workflow: scan card → draft email → send — in under 60 seconds.

3. Analysis Frameworks

3.1 SWOT Analysis

	Positive	Negative
Internal	Strengths: Local-first (privacy by design), on-device OCR (no cloud dependency), AI-powered drafts, iOS-native performance, zero backend cost for MVP	Weaknesses: iOS-only, Gemini API dependency, no offline AI, single-developer team
External	Opportunities: Growing sales-tech market, Apple ecosystem expansion (iPadOS/macOS), on-device LLMs maturing, potential B2B team features	Threats: Apple/Google could add similar features natively, Gemini API pricing changes, App Store review delays

3.2 PESTLE Analysis

Factor	Impact
Political	Low — consumer app, no regulated industry
Economic	Freemium model viable; Gemini Flash API cost is per-call and low
Social	Remote/hybrid work increases trade-show networking value
Technological	Swift 6, SwiftUI, Vision framework, on-device ML all mature
Legal	GDPR compliance required (data minimisation already built-in); App Store guidelines
Environmental	Minimal — local-first reduces cloud compute carbon footprint

4. Planning Phase

4.1 Problem Statement

Sales professionals lose lead quality between first contact at events and follow-up emails. Manual composition from business cards is slow, leading to cold, delayed outreach with lower conversion rates.

4.2 Project Objectives

#	Objective	Metric
O1	Reduce card-to-email time	< 60 seconds
O2	Achieve accurate OCR extraction	> 85% field accuracy
O3	Generate usable AI drafts	User edits < 20% of generated text
O4	Zero backend cost for MVP	\$0 infrastructure spend
O5	Ship MVP to TestFlight	Within 8 weeks

4.3 Project Charter

Item	Detail
Project Title	Reach Dog — AI-Powered Sales Lead CRM
Sponsor	Reach Dog Project Team
Start Date	April 2026
Target MVP	June 2026
Target Scalable	Q4 2026
Deliverables	iOS app (MVP), SRS document, test report

4.4 Project Scope

In-scope (MVP): Authentication (Apple/Google), OCR scan, lead CRUD, AI follow-up, email via URL schemes, CSV export/import, dashboard.

In-scope (Scalable): iCloud sync, Gmail API send, team workspaces, custom pipeline, reminders, advanced analytics.

Out-of-scope: Android, web, desktop, full email client, third-party CRM sync.

4.5 Stakeholder Analysis

Stakeholder	Role	Interest
Sales Agent	Primary user	Scan cards, generate emails, manage leads
Sales Manager	Scalable user	View team dashboard, monitor pipeline
Team Administrator	Scalable user	Manage team, shared templates, pipeline config

Stakeholder	Role	Interest
Developer	Builder	Technical implementation
App Store Reviewer	Gatekeeper	Compliance with Apple guidelines

4.6 User Characteristics

User Class	Technical Skill	Frequency	Platform
Sales Agent	Smartphone-proficient, no CRM training	Daily at events	iPhone, iOS 17+
Sales Manager	Moderate technical	Weekly review	iPhone, iOS 17+
Team Administrator	Moderate-to-high	Setup + periodic	iPhone, iOS 17+

4.7 Work Breakdown Structure

1.0 Reach Dog

1.1 Authentication

1.1.1 Apple Sign-In

1.1.2 Google Sign-In

1.1.3 Session management

1.2 OCR Module

1.2.1 Camera capture

1.2.2 Gallery import

1.2.3 Vision text recognition

1.2.4 Field parsing logic

1.3 Lead Management

1.3.1 SwiftData models

1.3.2 Lead list view

1.3.3 Lead profile view

1.3.4 Search & filter

1.3.5 Status management

1.4 AI Follow-Up

1.4.1 Gemini API integration

1.4.2 Prompt engineering

1.4.3 Template library

1.4.4 Draft review UI

1.5 Email Dispatch

1.5.1 URL scheme builder

1.5.2 Gmail scheme

1.5.3 Apple Mail scheme

1.5.4 Copy to clipboard

1.6 Settings & Data

1.6.1 Preferences screen

1.6.2 CSV export

1.6.3 CSV import

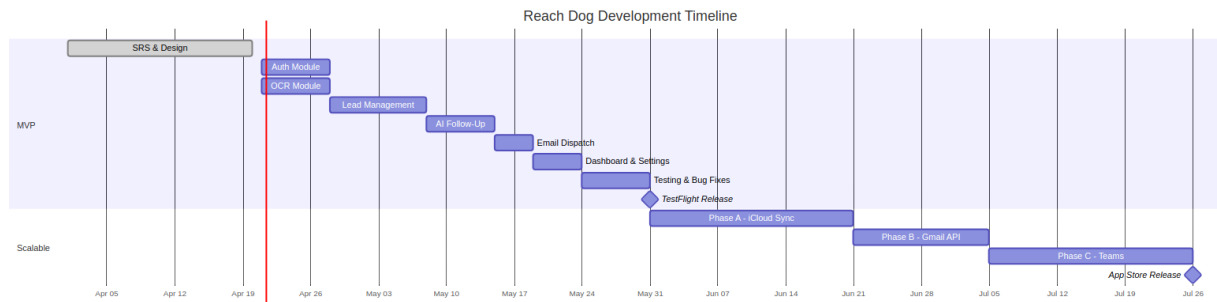
1.7 Dashboard

- 1.7.1 Status counts
- 1.7.2 Swift Charts visualisation
- 1.8 Testing & Deployment
 - 1.8.1 Unit tests
 - 1.8.2 Integration tests
 - 1.8.3 TestFlight submission

4.8 Time Management

Milestone	Target Date
SRS Complete	April 2026
Auth + OCR Module	Week 2
Lead Management	Week 4
AI + Email Dispatch	Week 5
Dashboard + Settings	Week 6
Testing + Bug Fixes	Week 7
MVP TestFlight	Week 8
Scalable Phase A (iCloud)	Week 14
Scalable Phase B (Gmail API)	Week 18
Scalable Phase C (Teams)	Week 24

4.9 Gantt Chart



4.10 Risk Management

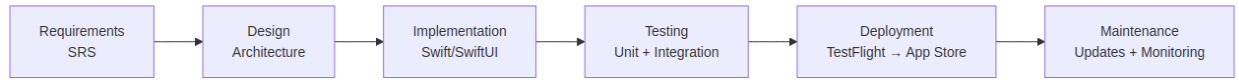
Risk	Likelihood	Impact	Mitigation
Gemini API deprecation/pricing change	Medium	High	Abstract AI service behind protocol; swap to on-device LLM
OCR accuracy on poor-quality cards	High	Medium	Allow manual field editing; iterative parser improvements
App Store rejection	Low	High	Follow HIG strictly; no private APIs
API key compromise	Low	Critical	Keychain storage; key rotation; rate limiting

Risk	Likelihood	Impact	Mitigation
Device theft exposing lead data	Medium	High	AES-256 at-rest encryption via Data Protection
CloudKit sync conflicts (Scalable)	Medium	Medium	Last-writer-wins with conflict UI

5. Methodology

5.1 Waterfall Methodology

This project follows a modified Waterfall approach: Requirements → Design → Implementation → Testing → Deployment. Each phase completes before the next begins, which suits a well-defined MVP with fixed scope. The scalable release follows the same phases incrementally (Phase A, B, C).



6. Analysis

6.1 Functional Requirements

Requirements marked **(MVP)** are prototype scope. **(Scalable)** are production evolution. *Shall* = binding.

6.1.1 Authentication and Identity

ID	Tier	Requirement
FR-1.1	MVP	The system shall offer Sign in with Apple via AuthenticationServices .
FR-1.2	MVP	The system shall offer Google Sign-In via GoogleSignIn-iOS SDK.
FR-1.3	MVP	The system shall persist a minimal user profile locally (provider ID, email, display name, provider label).
FR-1.4	MVP	The system shall provide a logout control that clears the session while retaining local data.
FR-1.5	Scalable	The system shall support team invitation via shareable deep link.
FR-1.6	Scalable	The system shall enforce role-based access control (Sales Agent, Sales Manager, Team Administrator).

6.1.2 Lead Capture via OCR

ID	Tier	Requirement
FR-2.1	MVP	The system shall present a Scan screen with live camera capture and photo-gallery import.
FR-2.2	MVP	The system shall invoke VNRecognizeTextRequest in fast mode on acquired images.
FR-2.3	MVP	The system shall parse recognised text into Full Name, Company, Job Title, Email, Phone, and Website fields.

ID	Tier	Requirement
FR-2.4	MVP	The system shall present a Review Contact screen with the card image and editable extracted fields.
FR-2.5	MVP	The system shall permit free-form tag attachment during review.
FR-2.6	Scalable	The system shall permit batch capture (up to 10 cards before review).
FR-2.7	Scalable	The system shall deduplicate against existing records via enterprise contact SDKs.

6.1.3 Lead Management

ID	Tier	Requirement
FR-3.1	MVP	The system shall persist leads in the local SwiftData store.
FR-3.2	MVP	The system shall render leads as a scrollable card-based list.
FR-3.3	MVP	The system shall provide case-insensitive substring search over name, company, and email.
FR-3.4	MVP	The system shall open a Lead Profile view on card selection.
FR-3.5	MVP	The Lead Profile shall support status editing among new , follow-up , and contacted .
FR-3.6	MVP	The system shall permit lead deletion with confirmation.
FR-3.7	Scalable	The system shall support user-defined pipeline stages.
FR-3.8	Scalable	The system shall present a Kanban view grouped by pipeline stage.

6.1.4 AI Follow-Up Generation

ID	Tier	Requirement
FR-4.1	MVP	The Lead Profile shall expose a Generate Follow-Up control.

ID	Tier	Requirement
FR-4.2	MVP	The system shall compose a prompt (name, company, title, notes) and dispatch to Gemini Flash API.
FR-4.3	MVP	The system shall present one or more draft variations with subject and body.
FR-4.4	MVP	The system shall offer a library of static templates parameterised with lead fields.
FR-4.5	MVP	The system shall instruct the AI to limit bodies to ~300 words.
FR-4.6	Scalable	The system shall support user-authored prompt templates.
FR-4.7	Scalable	The system shall cache the most recent generation per lead for offline review.

6.1.5 Email Dispatch

ID	Tier	Requirement
FR-5.1	MVP	The system shall provide a Copy control (system pasteboard).
FR-5.2	MVP	The system shall construct an email URL with pre-filled to, subject, body.
FR-5.3	MVP	Gmail: <code>googlegmail://co?</code> scheme.
FR-5.4	MVP	Apple Mail: <code>mailto:</code> scheme (default).
FR-5.5	MVP	Mark Sent → status updated to contacted .
FR-5.6	Scalable	Native in-app Gmail API send (OAuth scope: <code>gmail.send</code>).
FR-5.7	Scalable	Native in-app Apple Mail via <code>MFMailComposeViewController</code> .
FR-5.8	Scalable	Delivery confirmation tracking where available.

6.1.6 Settings and Data Portability

ID	Tier	Requirement
FR-6.1	MVP	Settings: preferred mail client selector (Gmail / Apple Mail) + logout.
FR-6.2	MVP	CSV export via iOS share sheet.
FR-6.3	MVP	CSV import, appending to existing store.
FR-6.4	Scalable	iCloud sync via CloudKit private database.
FR-6.5	Scalable	Scheduled local reminders for follow-up leads.

6.1.7 Analytics Dashboard

ID	Tier	Requirement
FR-7.1	MVP	Home dashboard: lead counts per status + Swift Charts visualisation.
FR-7.2	Scalable	Time-series: leads/week, conversion rate.
FR-7.3	Scalable	Sales Manager aggregated team dashboard.

6.1.8 Team Collaboration (Scalable)

ID	Tier	Requirement
FR-8.1	Scalable	Team creation → creator becomes Team Administrator.
FR-8.2	Scalable	Invite members via time-limited link.
FR-8.3	Scalable	Shared template library (Team Admin editable).
FR-8.4	Scalable	Lead ownership transfer between team members.

6.2 Non-Functional Requirements

6.2.1 Performance

ID	Tier	Requirement	Target
NFR-P1	MVP	OCR extraction latency	2 s per card
NFR-P2	MVP	Peak resident memory	400 MB

ID	Tier	Requirement	Target
NFR-P3	MVP	Sustained CPU utilisation	30%
NFR-P4	MVP	Local storage footprint	500 MB
NFR-P5	MVP	Binary size	300 MB
NFR-P6	MVP	AI round-trip latency	5 s on 4G
NFR-P7	Scalable	CloudKit sync convergence	15 s
NFR-P8	Scalable	Concurrent team users	50

6.2.2 Security

ID	Requirement
NFR-S1	AES-256 at-rest encryption via <code>FileProtectionType.complete</code>
NFR-S2	Gemini API key stored in Keychain (<code>kSecAttrAccessibleWhenUnlockedThisDeviceOnly</code>)
NFR-S3	All network traffic over TLS 1.2+ with ATS enabled
NFR-S4	Identity tokens validated within Apple/Google SDKs only
NFR-S5	No PII logged in production builds
NFR-S6	CloudKit private containers per user (Scalable)

6.2.3 Privacy

- Only name, company, title, notes sent to Gemini API — minimum necessary for draft generation
- Business-card images never transmitted externally
- GDPR Article 5 compliant: data minimisation, purpose limitation, user control
- No lead data stored on company infrastructure (MVP)

6.2.4 Usability

ID	Requirement
NFR-U1	Conforms to Apple HIG
NFR-U2	Dynamic Type across all primary text
NFR-U3	Passes Xcode Accessibility Audit
NFR-U4	Scan → email sent in 6 taps

6.2.5 Reliability

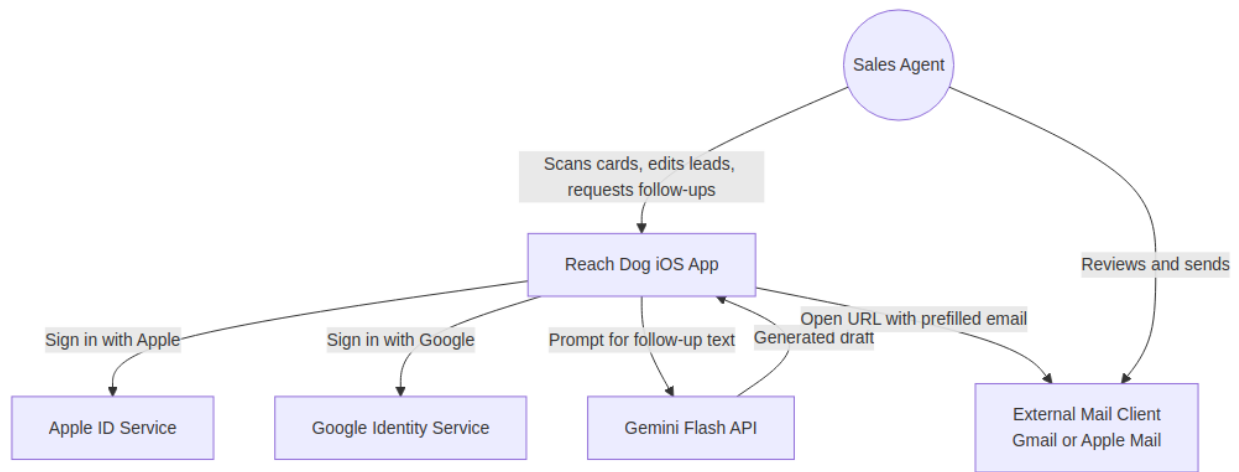
- MTBF > 500 hours active use
- Graceful degradation: static templates when Gemini is unreachable

6.2.6 Maintainability

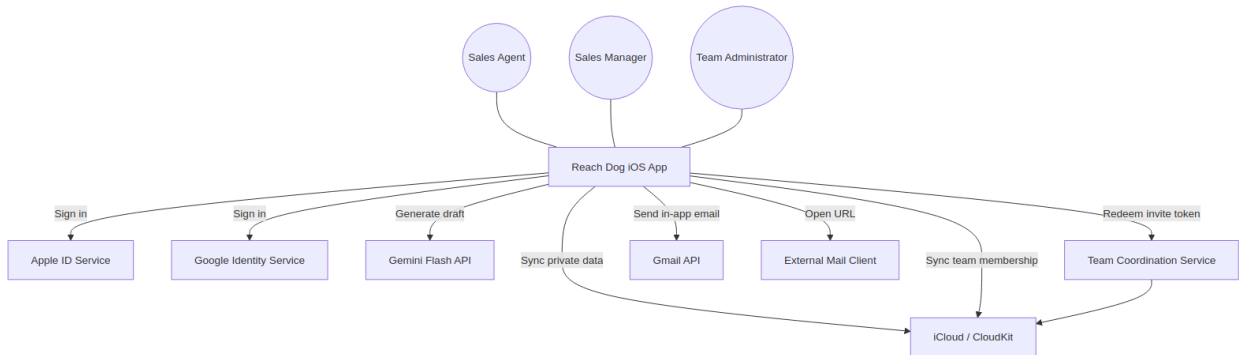
- MVVM pattern throughout
- 70% unit-test coverage for domain services
- Swift concurrency (`async/await`) for all async operations

7. Process and Data Modelling

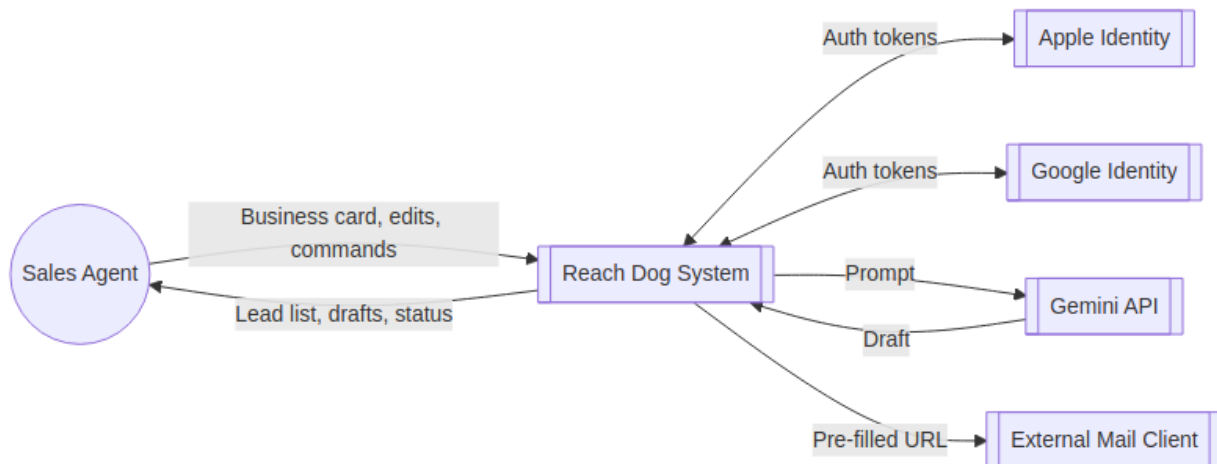
7.1 Context Diagram (MVP)



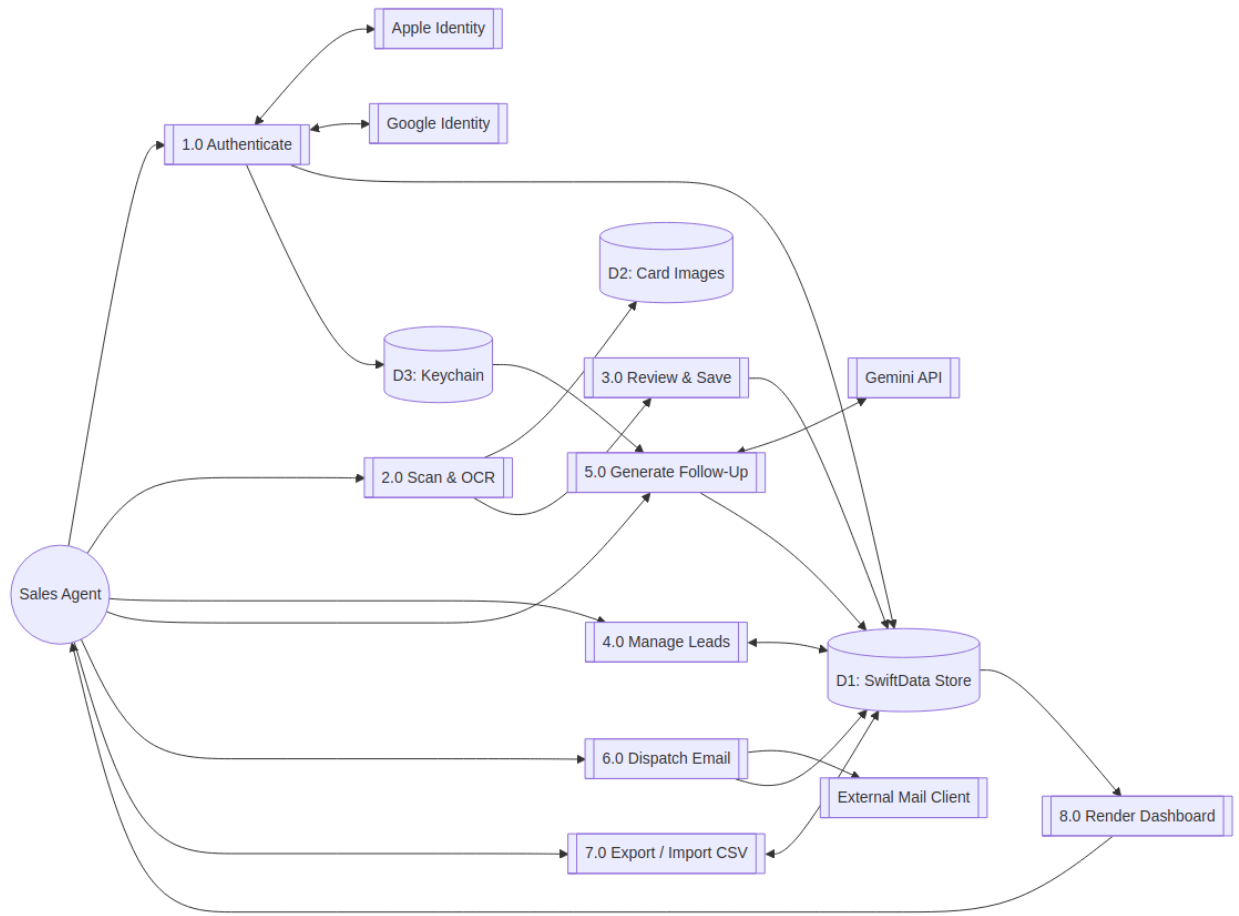
7.2 Context Diagram (Scalable)



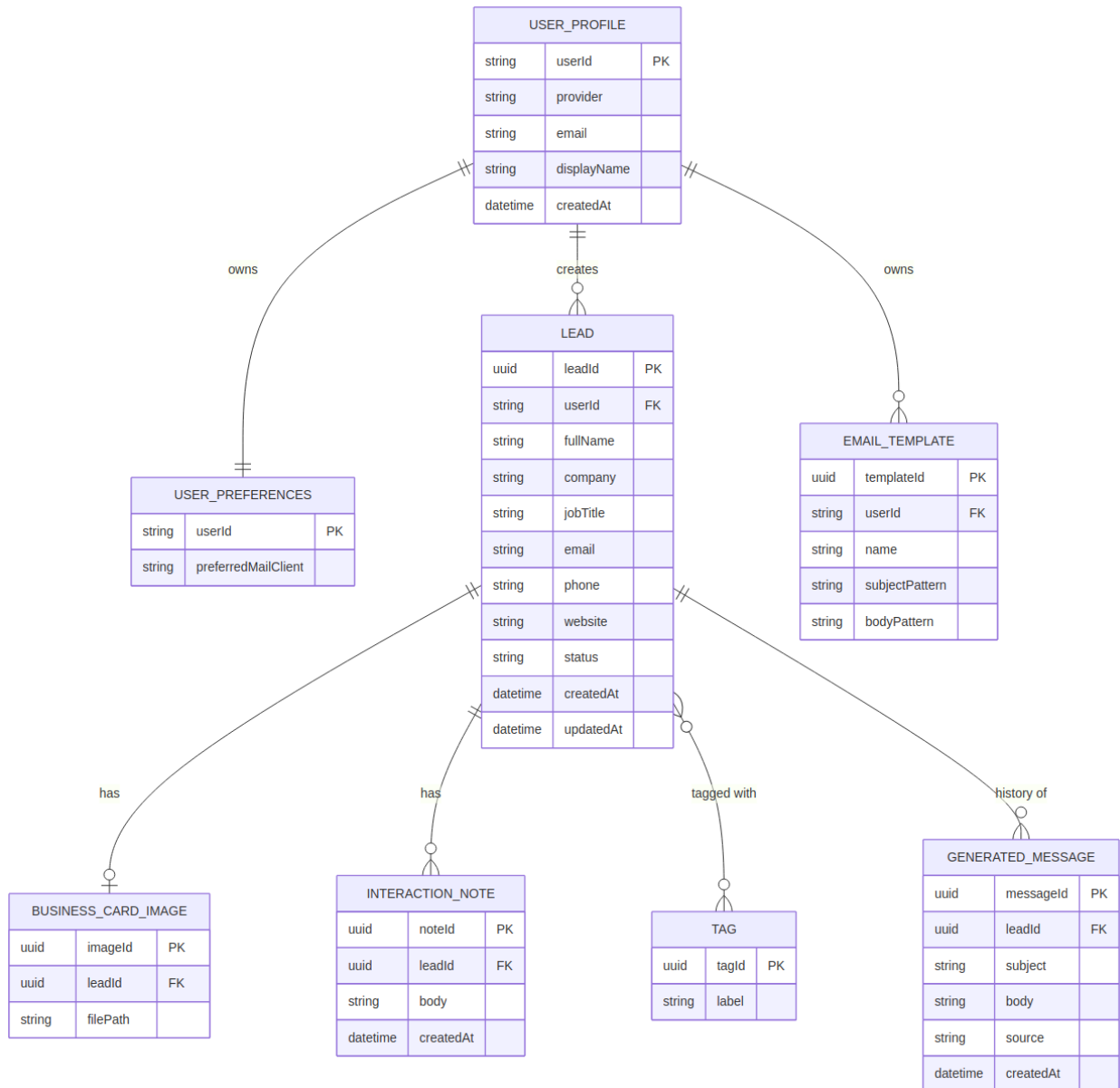
7.3 DFD Level 0



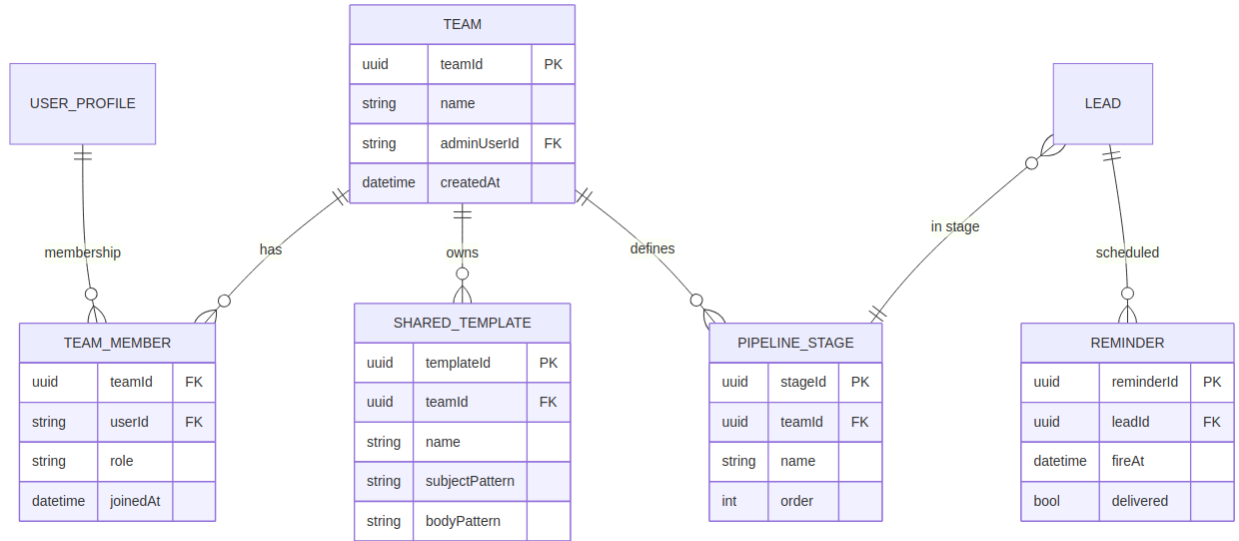
7.4 DFD Level 1 (MVP)



7.5 Entity-Relationship Diagram (MVP)



7.6 Entity-Relationship Diagram (Scalable Extensions)



7.7 Database Schema (SwiftData)

UserProfile

Field	Type	Nullable	Notes
userId	String (PK)	No	Provider-assigned
provider	String	No	apple / google
email	String	No	
displayName	String	Yes	
createdAt	Date	No	

UserPreferences

Field	Type	Nullable	Notes
userId	String (PK, FK)	No	1:1 with UserProfile
preferredMailClient	String	No	gmail / appleMail
reminderDefaultInDays	Int	Yes	Scalable only

Lead

Field	Type	Nullable	Notes
leadId	UUID (PK)	No	
userId	String (FK)	No	
fullName	String	No	
company	String	Yes	
jobTitle	String	Yes	
email	String	Yes	

Field	Type	Nullable	Notes
phone	String	Yes	
website	String	Yes	
status	String	No	Enum: new , follow-up , contacted
stageId	UUID (FK)	Yes	Scalable only
createdAt	Date	No	
updatedAt	Date	No	

BusinessCardImage

Field	Type	Nullable	Notes
imageId	UUID (PK)	No	
leadId	UUID (FK)	No	
filePath	String	No	Sandbox-relative path

InteractionNote

Field	Type	Nullable	Notes
noteId	UUID (PK)	No	
leadId	UUID (FK)	No	
body	String	No	
createdAt	Date	No	

Tag *(many-to-many via implicit join set)*

Field	Type	Nullable	Notes
tagId	UUID (PK)	No	
label	String	No	

GeneratedMessage

Field	Type	Nullable	Notes
messageId	UUID (PK)	No	
leadId	UUID (FK)	No	
subject	String	No	
body	String	No	
source	String	No	ai / template
createdAt	Date	No	

EmailTemplate

Field	Type	Nullable	Notes
templateId	UUID (PK)	No	
userId	String (FK)	No	
name	String	No	
subjectPattern	String	No	Tokens: {fullName}, {company}, etc.
bodyPattern	String	No	

8. System Design

8.1 Architectural Overview (MVP)

Three-layer client-only architecture on iOS:

1. **Presentation Layer** — SwiftUI views + MVVM ViewModels
2. **Domain Layer** — Stateless services: `AuthService`, `OCRService`, `LeadService`, `AIService`, `EmailDispatcher`, `CSVService`
3. **Persistence Layer** — `SwiftData` (structured data), File System (card images), Keychain (API key, tokens)

No backend. External dependencies limited to three HTTPS services: Apple Identity, Google Identity, Gemini Flash API.

Key properties: Zero operational cost, strong privacy by construction (data never leaves device except Gemini prompt), fully functional offline (except AI generation).

8.2 Architectural Overview (Scalable)

Preserves local-first. Adds: - **CloudKit** — per-user private sync across devices (user’s own iCloud, not company infra) - **Serverless coordination service** (AWS Lambda / Cloudflare Workers) — team invitation tokens and membership arbitration only. No lead data passes through it.

8.2.1 Key Design Rationales

Several design choices in this specification trade off richness for simplicity, and are documented here so reviewers can evaluate them explicitly.

Company stored as plain text on Lead, not as a separate entity. A Company entity with autocomplete reuse across leads would enrich the CRM, but it introduces non-trivial UX complexity (fuzzy matching, de-duplication, merge flows). The MVP’s scanning workflow already fills the Company field directly from the business card via OCR; if OCR fails, the user types it freely. Promoting Company to an entity is deferred to a scalable-release enhancement and is **out of MVP scope**.

“Event” concept dropped in favour of free-form tags. An earlier design iteration considered a separate `Event` entity to track which conference or meeting a lead was captured at. On review of the captured-UI screens, no `Event` field surfaced to the user; tags (e.g. `#techconf-2026`) cover the same informational need with zero schema cost. Event is therefore **explicitly excluded from both MVP and scalable releases**, and tags are the canonical mechanism for grouping leads by origin.

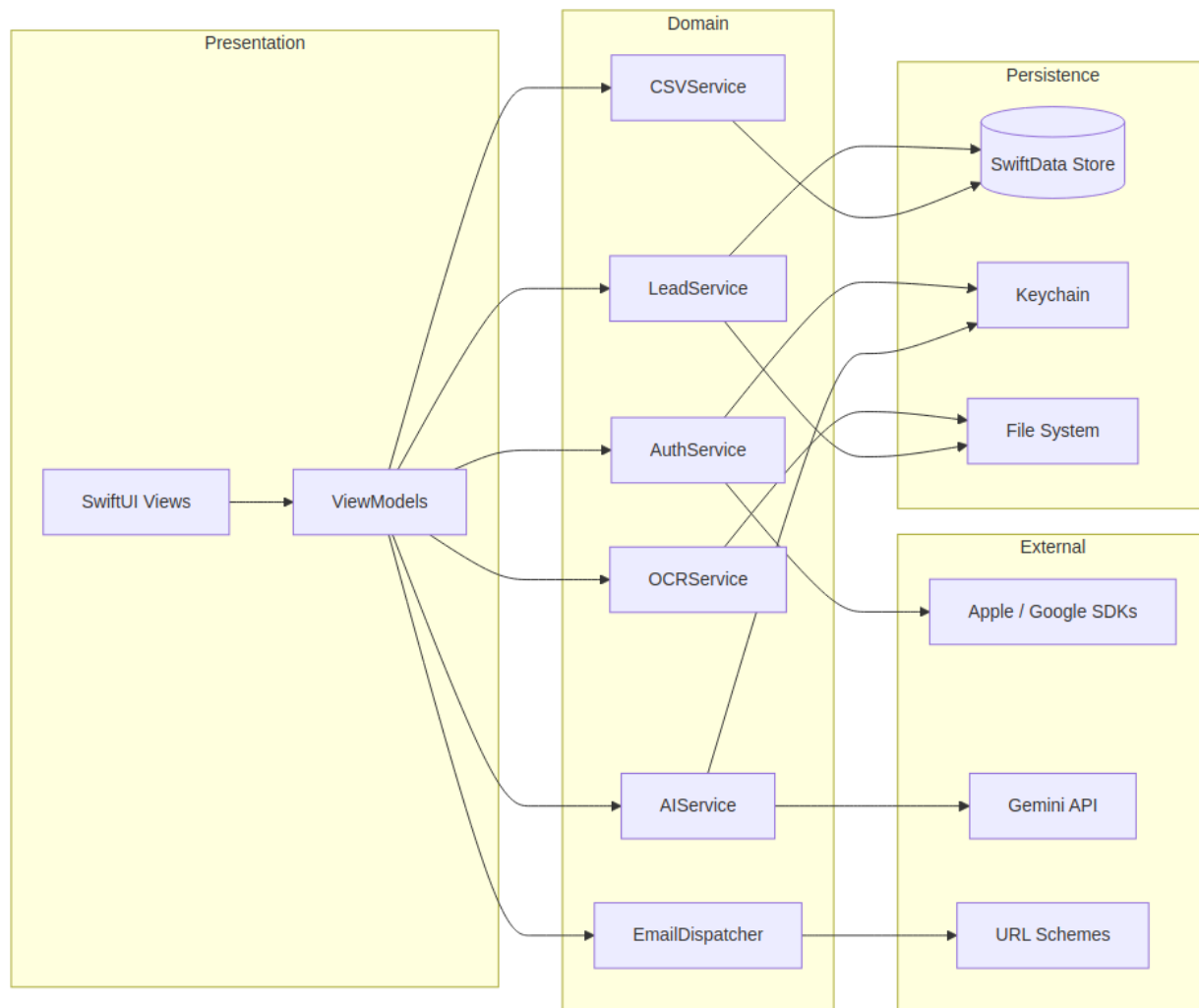
CSV export/import is the MVP’s cross-device migration strategy. Since MVP data is local-only (no CloudKit in MVP, no backend), users switching devices export their data as CSV and re-import on the new device. This is not a limitation to apologise for; it is a deliberate **privacy-preserving ownership** choice consistent with the “local-first” principle. Cross-device automatic sync is introduced in Scalable Phase A via CloudKit private containers (FR-6.4).

AI-generated email bodies are capped at ~300 words (FR-4.5) to stay within URL-scheme length tolerances (FR-5.2). The MVP dispatches email by building a `mailto:` or `googlegmail://` URL with pre-filled subject and body. Observed truncation thresholds on iOS

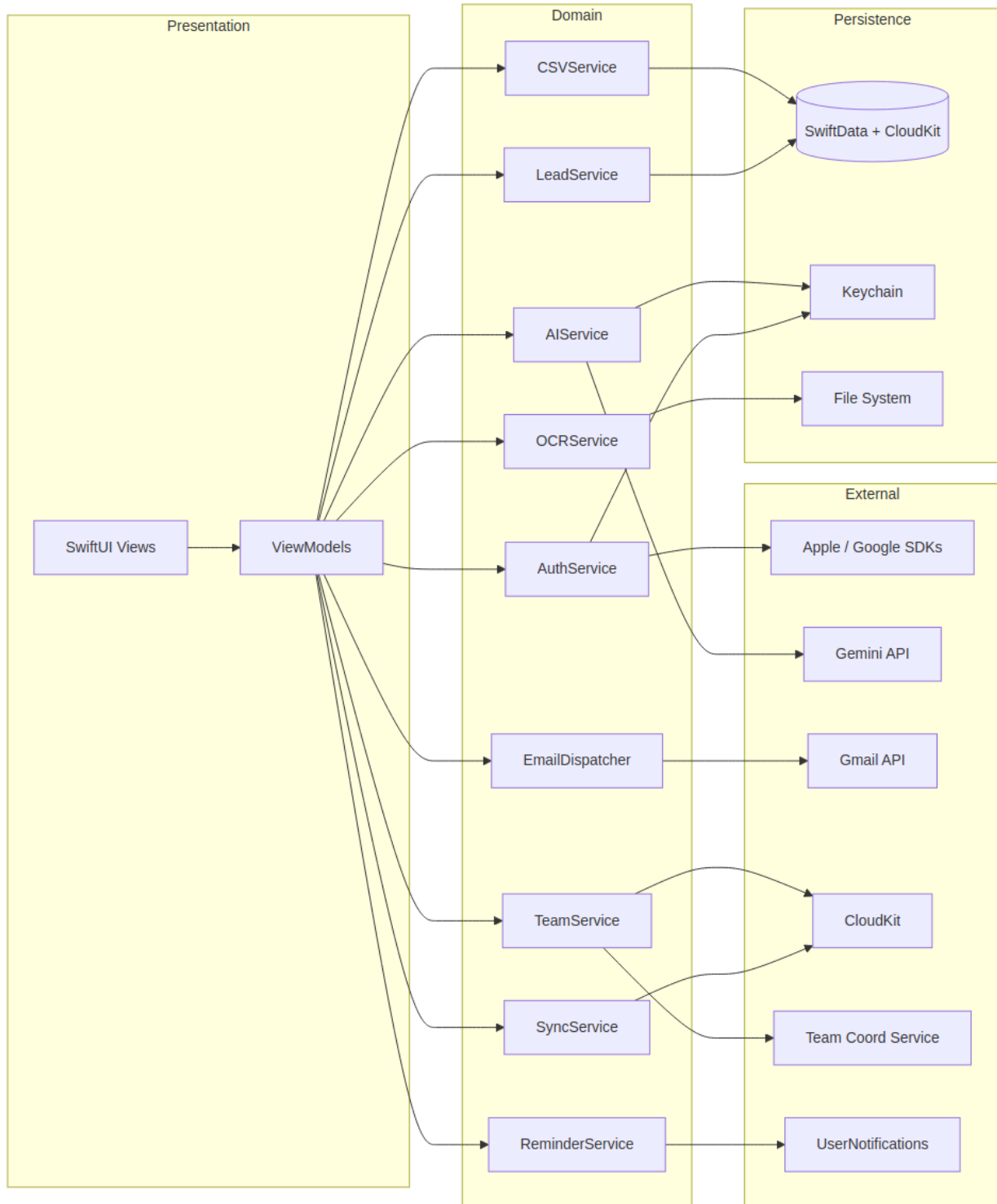
are approximately 2000–4000 characters, and a 300-word email is roughly 1500–2000 characters — comfortably inside the safe margin. FR-4.5 is therefore not an arbitrary quality constraint; it is a **technical prerequisite for FR-5.2 to operate reliably**.

Auto-fill both subject and body (vs. copy-then-paste). An earlier design considered having the user copy the body to the clipboard and paste it into an empty compose window. The final design auto-fills both fields, reducing the flow to a single tap, at the cost of being sensitive to URL length (mitigated by FR-4.5 above). The Copy button is retained as a secondary action for users who wish to paste the draft into a non-email destination such as LinkedIn messaging or WhatsApp.

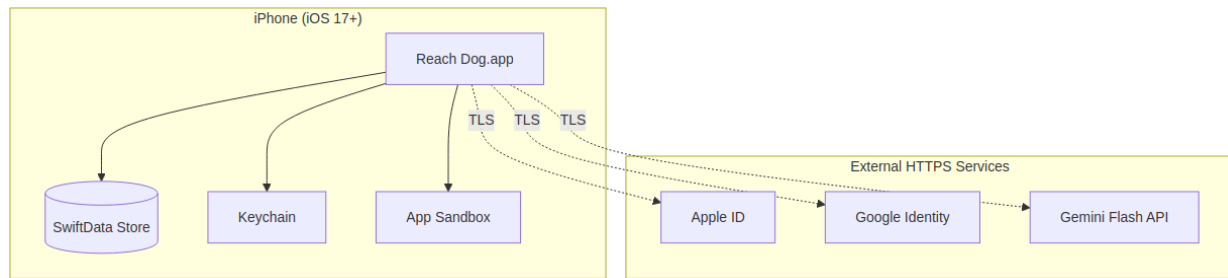
8.3 Component Diagram (MVP)



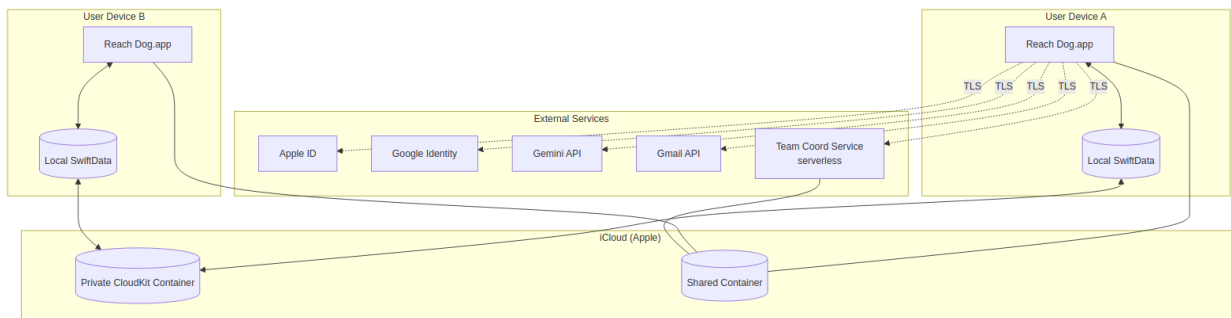
8.4 Component Diagram (Scalable)



8.5 Deployment Diagram (MVP)



8.6 Deployment Diagram (Scalable)



8.7 External Service Interfaces

8.7.1 Gemini Flash API

Endpoint: POST <https://generativelanguage.googleapis.com/v1/models/gemini-flash:generateContent>

Request:

```
{
  "contents": [{
    "parts": [{
      "text": "Draft a concise professional follow-up email to {fullName} ({jobTitle} at {comp"
    }]
  }],
  "generationConfig": { "maxOutputTokens": 512, "temperature": 0.7 }
}
```

Response:

```
{
  "candidates": [{
    "content": { "parts": [{ "text": "Subject: ...\n\nHi {fullName},\n\n..." }] }
  ]
}
```

8.7.2 Gmail API (Scalable)

Endpoint: POST <https://gmail.googleapis.com/gmail/v1/users/me/messages/send> **Auth:** OAuth 2.0 bearer token, scope <https://www.googleapis.com/auth/gmail.send> **Body:** base64url-encoded RFC 2822 message

8.7.3 URL Schemes (MVP)

Scheme	Client	Format
mailto:	Apple Mail	mailto:{to}?subject={s}&body={b}
googlegmail://	Gmail	googlegmail://co?to={to}&subject={s}&b

8.8 Internal Service Operations

Service	Operation	Purpose
AuthService	signInWithApple()	Initiate Apple auth flow
AuthService	signInWithGoogle()	Initiate Google auth flow
AuthService	signOut()	Terminate session
OCRService	recognise(image: UIImage) -> OCRResult	Extract text from card
LeadService	create(lead: LeadDraft) -> Lead	Persist new lead
LeadService	list(filter: LeadFilter) -> [Lead]	Retrieve leads
LeadService	update(lead: Lead) -> Lead	Persist edits
LeadService	delete(leadId: UUID)	Remove lead
AIService	generateFollowUp(lead: Lead, notes: String) -> [Message]	Query Gemini
EmailDispatcher	dispatch(message: Message, to: Lead)	Open external client
CSVService	exportAll() -> URL	Produce CSV file
CSVService	import(from: URL) -> ImportReport	Ingest CSV file

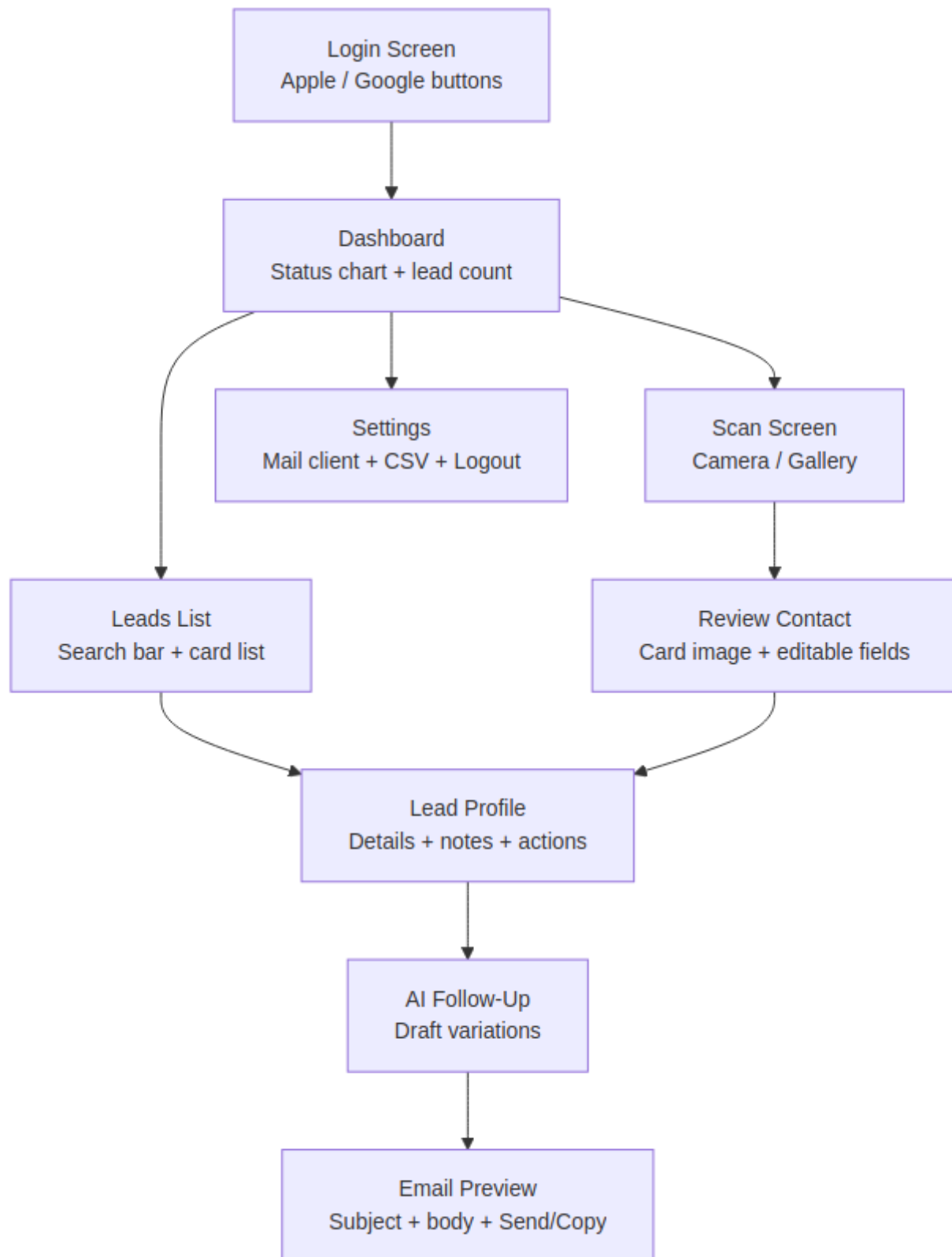
Scalable additions:

Service	Operation	Purpose
SyncService	pullChanges() / pushChanges()	CloudKit sync
TeamService	createTeam(name:)	Bootstrap a team
TeamService	generateInvite(teamId:)	Mint invitation token
TeamService	redeemInvite(token:)	Join a team

Service	Operation	Purpose
ReminderService	<code>schedule(lead: Lead, at: Date)</code>	Register local notification

8.9 UI Design — Key Screens (MVP)

Screen Map

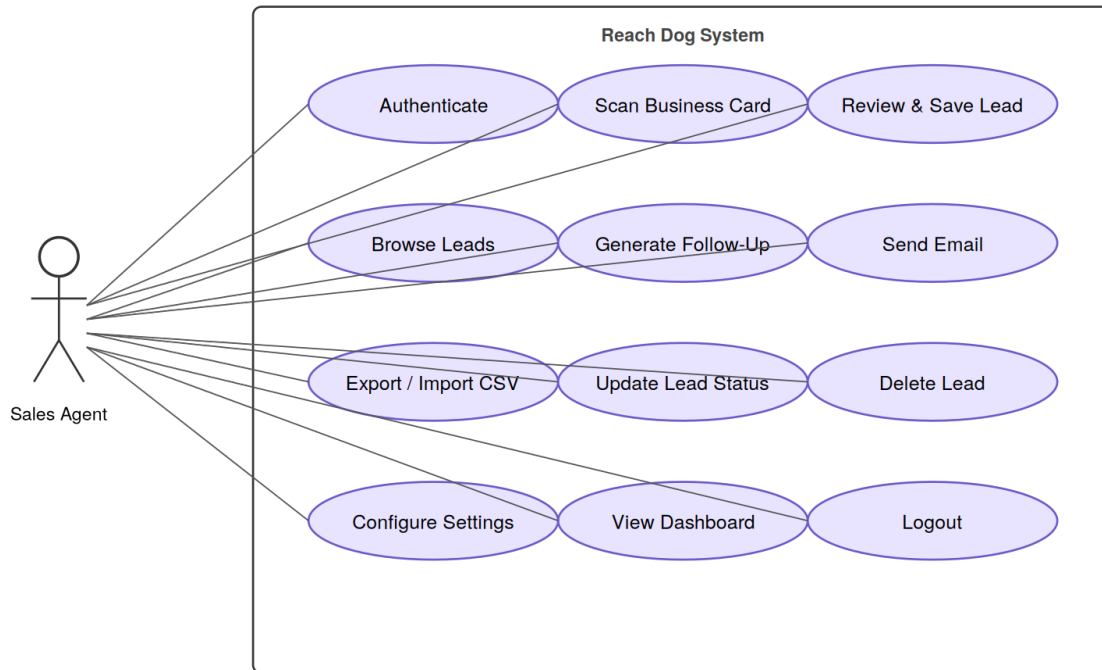


Screen Descriptions

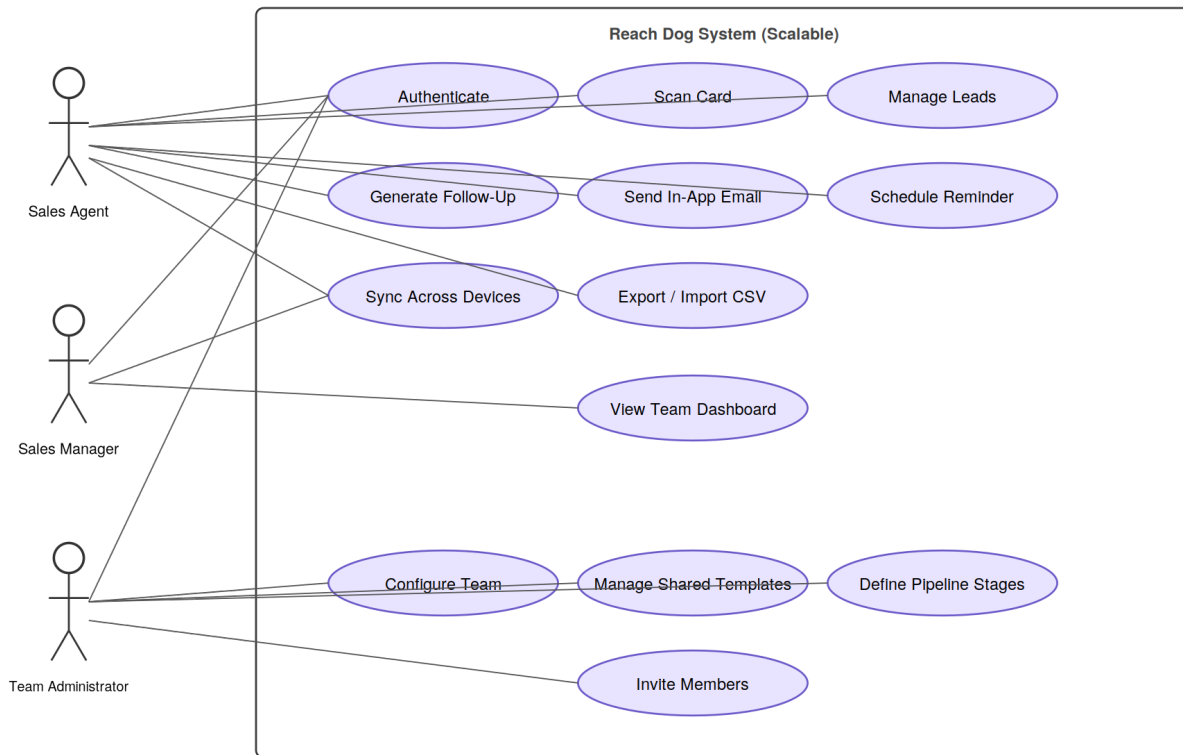
Screen	Layout	Key Elements
Login	Centred logo + 2 buttons	Apple Sign-In button (system style), Google Sign-In button
Dashboard	Top: status chart (Swift Charts pie/bar), Bottom: recent leads	Lead count badges (New / Follow-Up / Contacted)
Leads List	Top: search bar, Body: scrollable card list	Each card: name, company, status badge, avatar placeholder
Scan	Full-screen camera viewfinder	Shutter button, gallery icon, flash toggle
Review Contact	Top: card image thumbnail, Body: form fields	Fields: Name, Company, Title, Email, Phone, Website, Tags
Lead Profile	Header: name + company, Body: details + notes list	Action buttons: Generate Follow-Up, Edit, Delete
AI Follow-Up	Segmented: AI / Templates tab	Draft cards with subject + body preview, Select button
Email Preview	Subject field, body text area	Buttons: Send via Gmail, Send via Mail, Copy, Cancel
Settings	Grouped list	Mail client picker, Export CSV, Import CSV, Logout

9. UML Diagrams

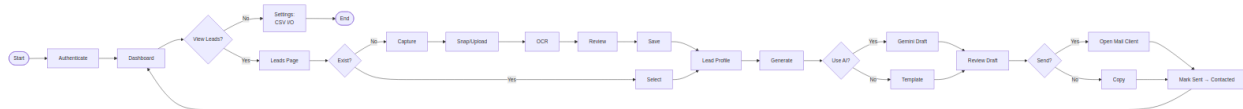
9.1 Use Case Diagram (MVP)



9.2 Use Case Diagram (Scalable)

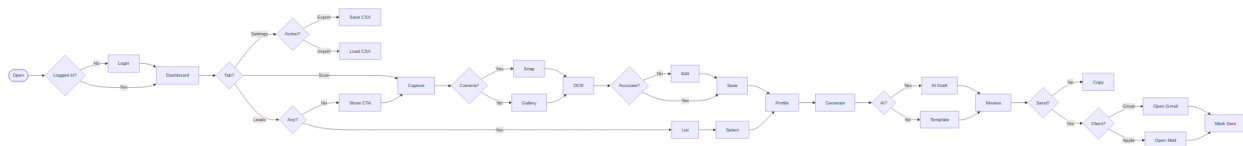


9.3 Activity Diagram — Primary Flow (MVP)

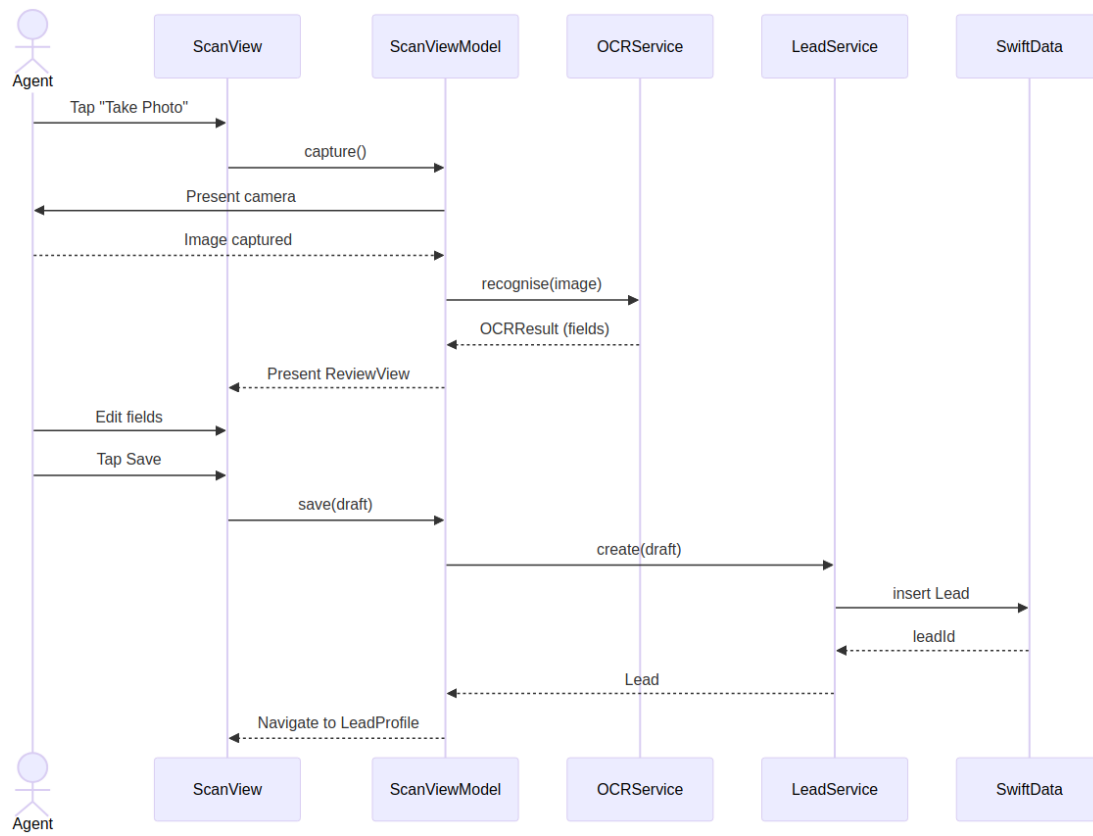


9.3.1 Flowchart — Detailed Decision Tree (MVP)

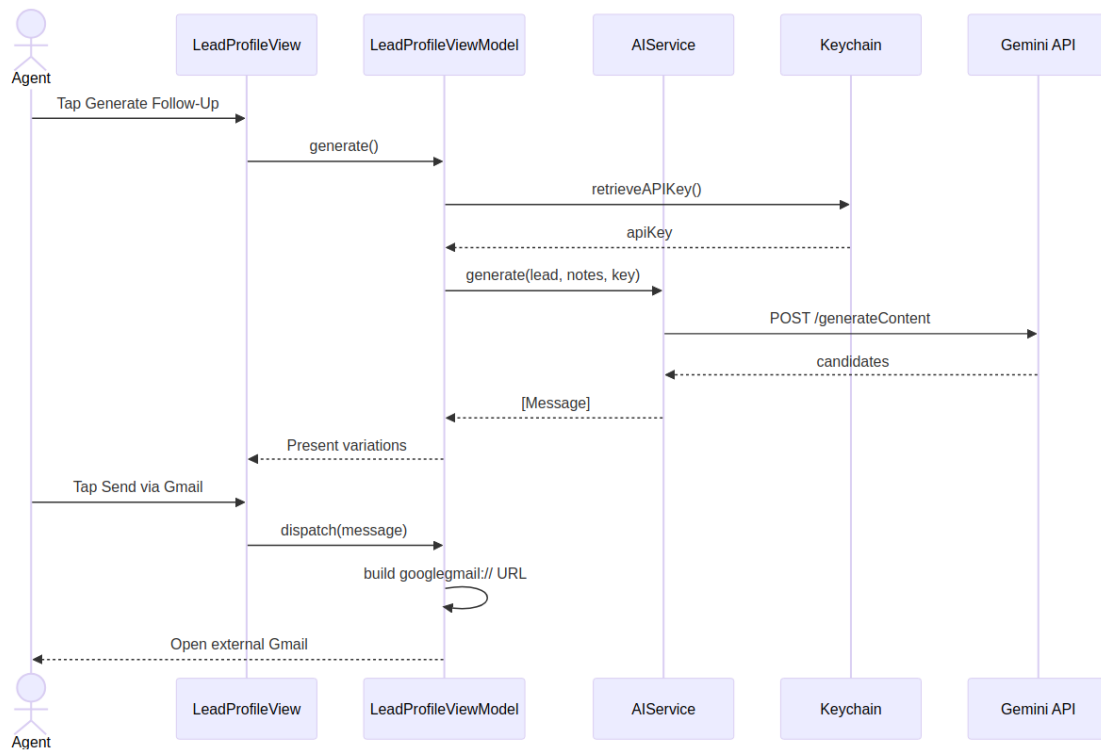
The flowchart complements the activity diagram by exposing every boolean decision point in the user flow. Each diamond corresponds to an `if` statement in the application logic.



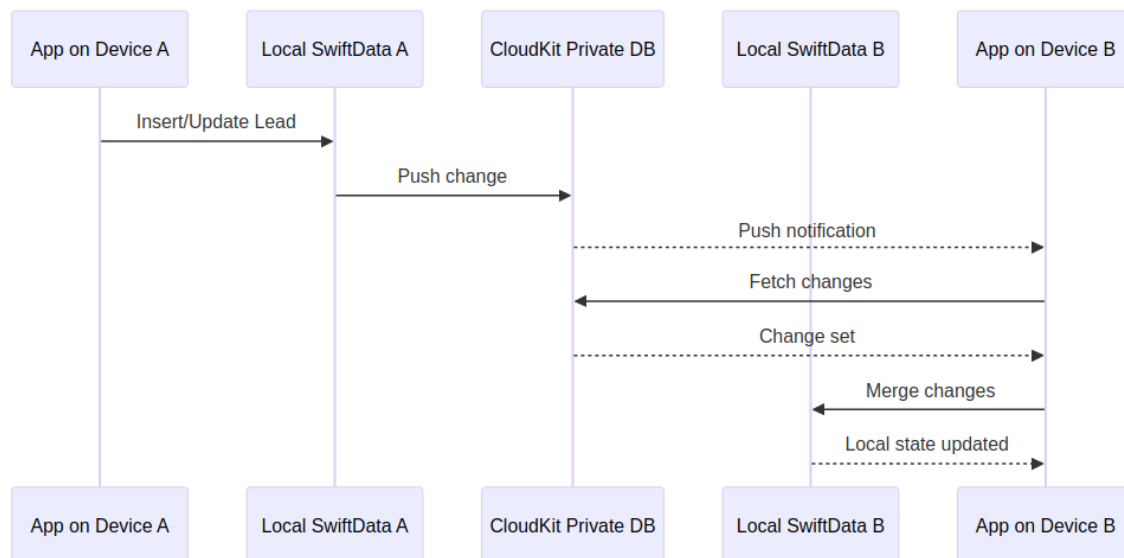
9.4 Sequence Diagram — Scan and Save



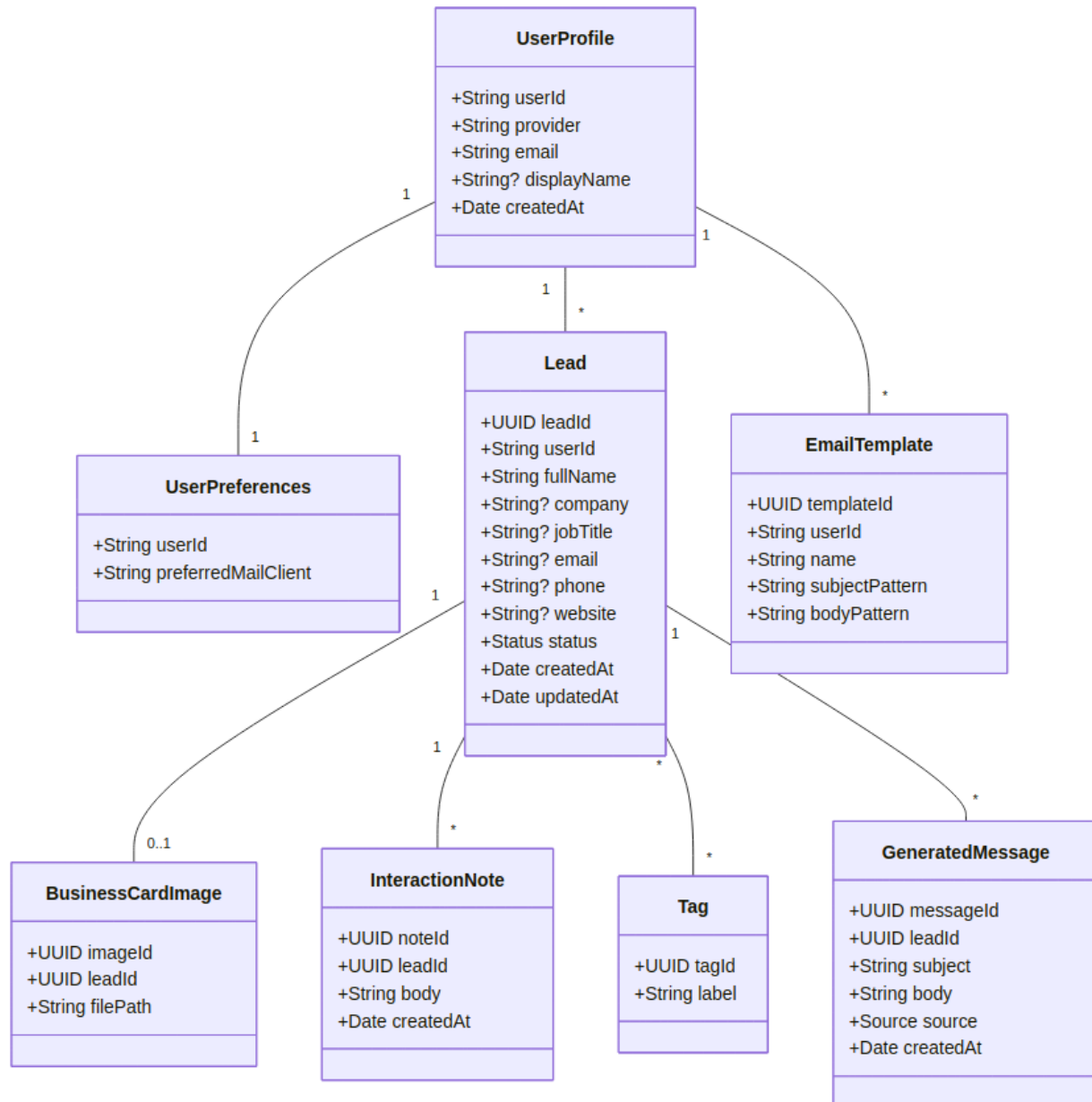
9.5 Sequence Diagram — Generate AI Follow-Up



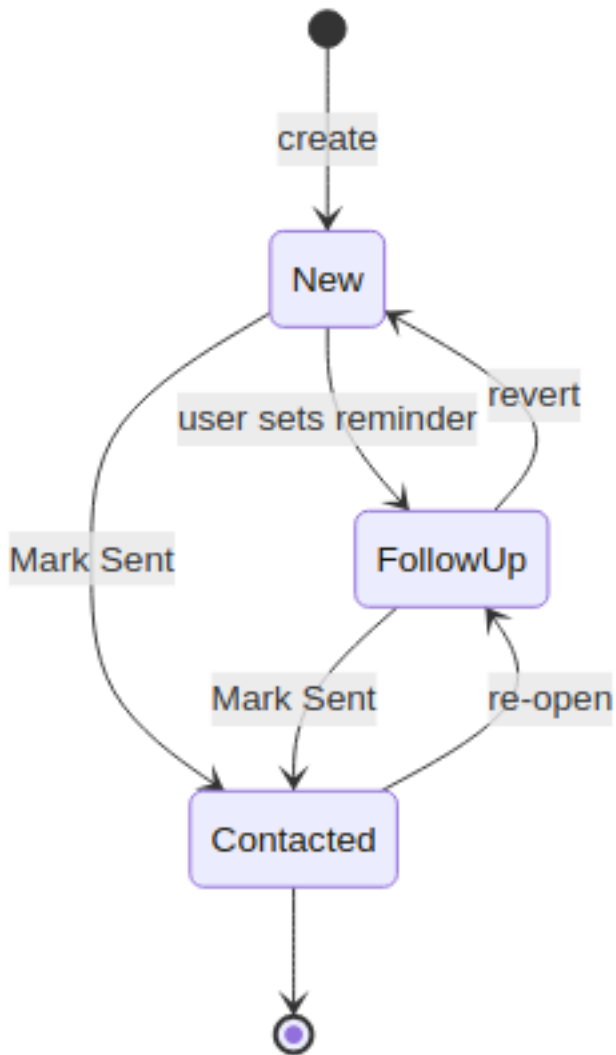
9.6 Sequence Diagram — CloudKit Sync (Scalable)



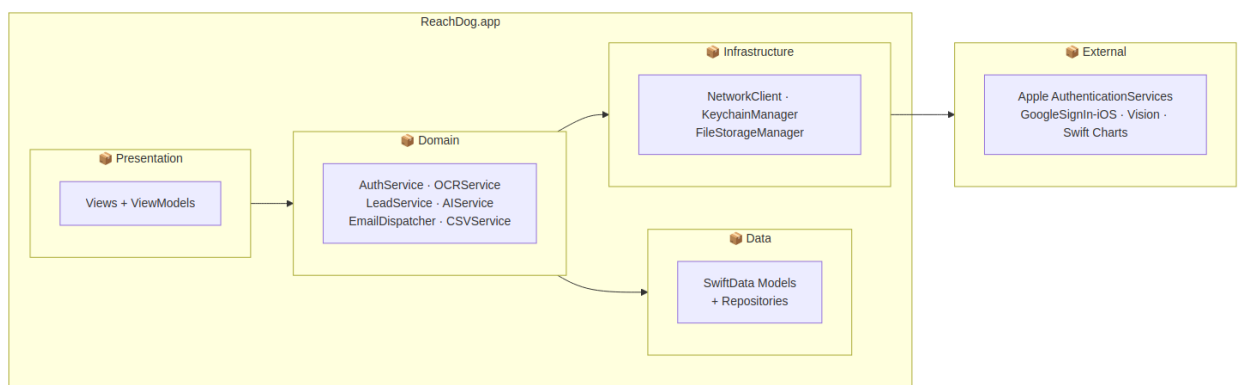
9.7 Class Diagram — Domain Model (MVP)



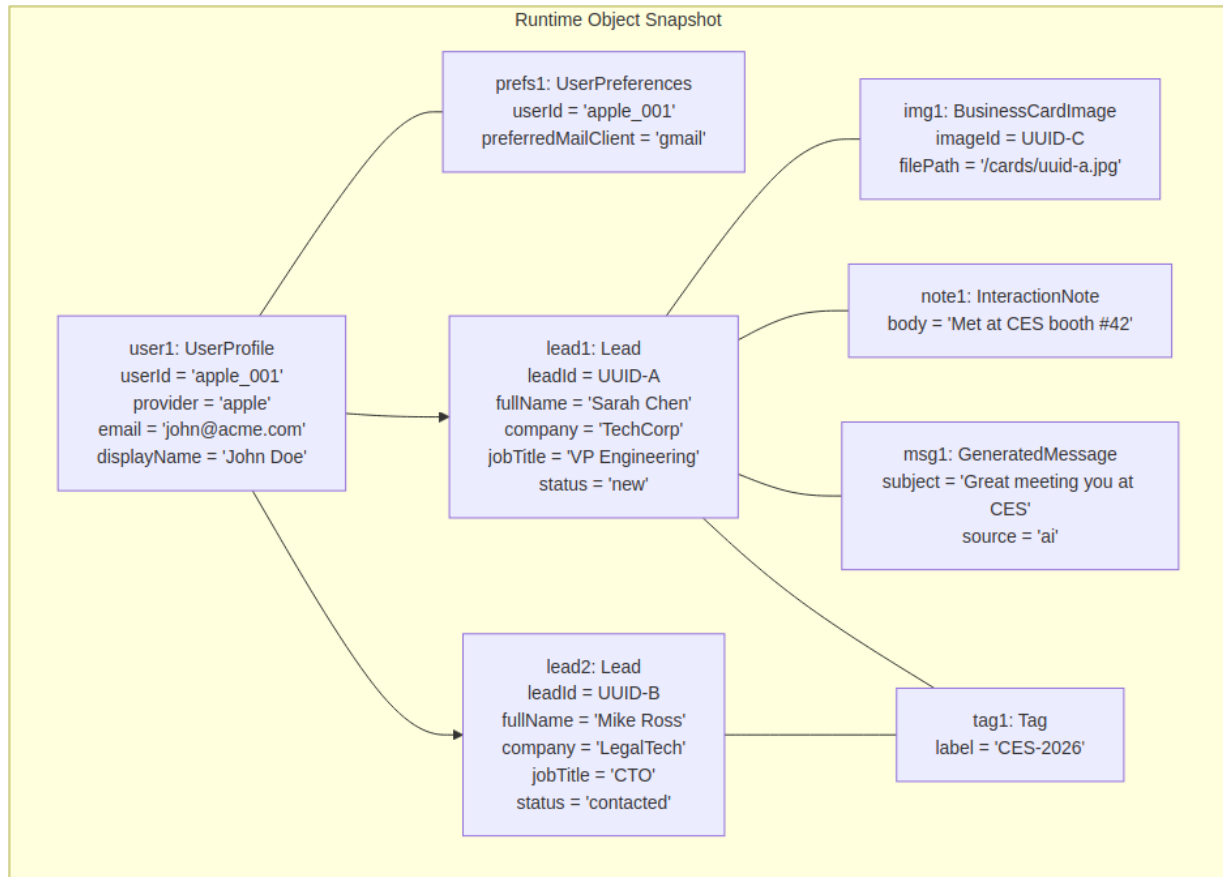
9.8 State Machine Diagram — Lead Lifecycle



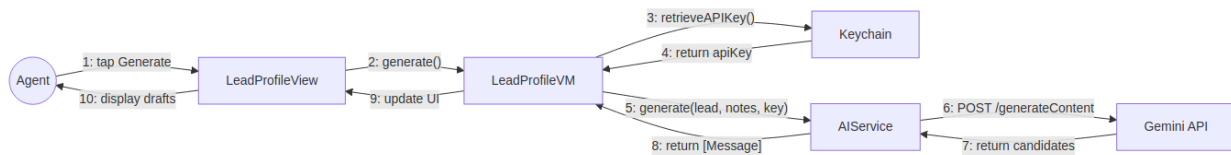
9.9 Package Diagram



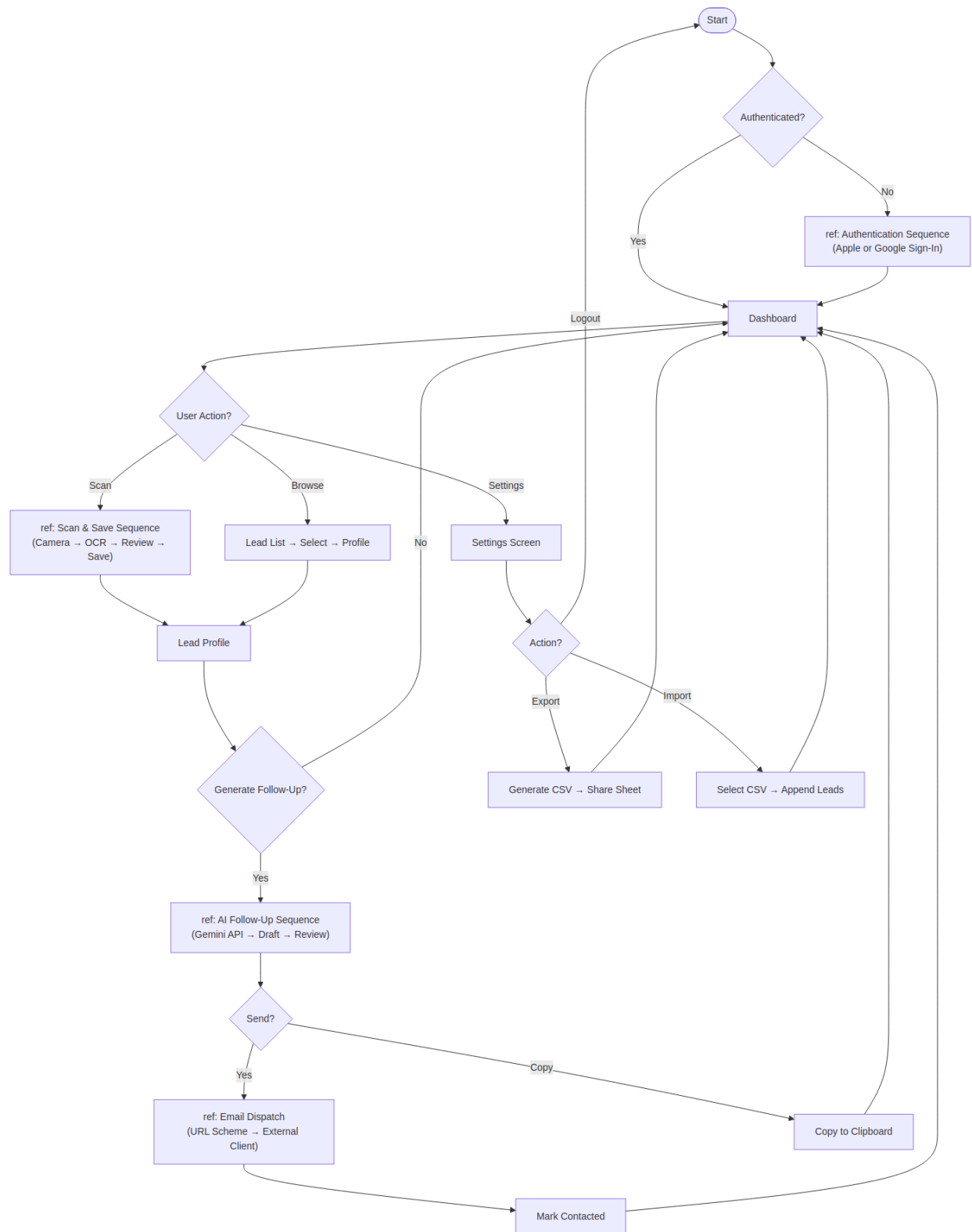
9.10 Object Diagram — Runtime Snapshot



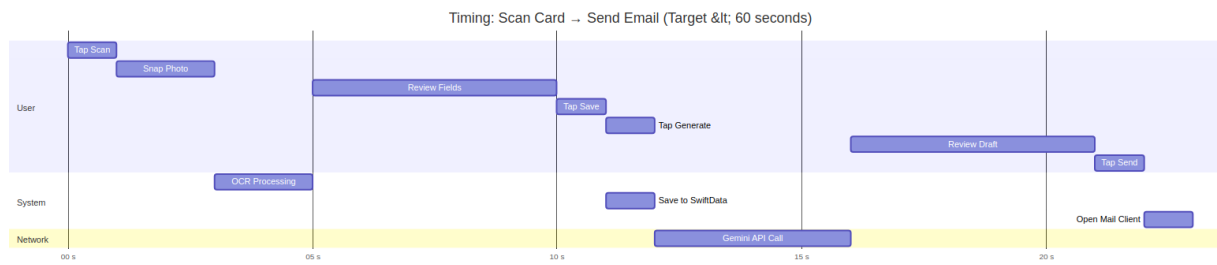
9.11 Communication Diagram — Generate Follow-Up



9.12 Interaction Overview Diagram — Main App Flow



9.13 Timing Diagram — Scan-to-Email Flow



Total estimated time: ~23 seconds — well within the 60-second target.

10. MVP Design

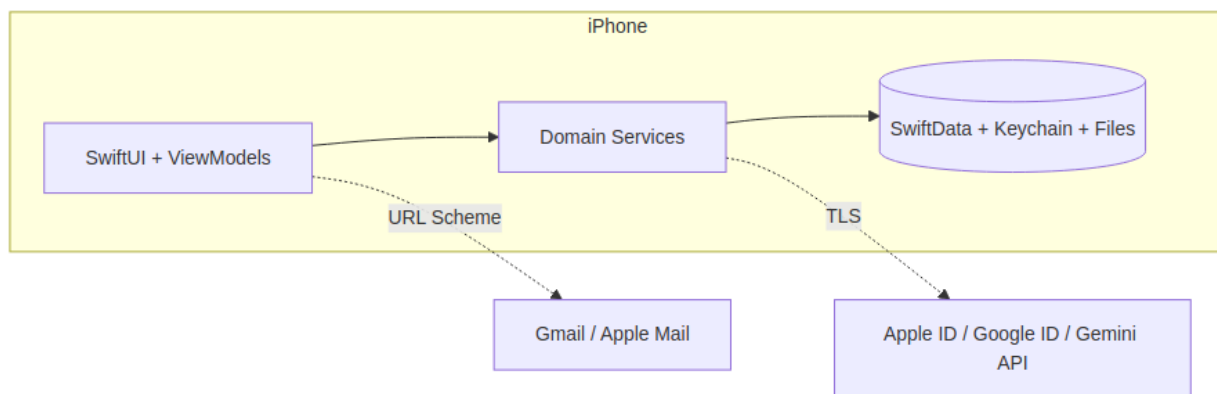
10.1 MVP Definition

The MVP is a fully functional iOS application that delivers the core value proposition: **scan a business card and have a personalised follow-up email ready to send within 60 seconds**. It runs entirely on-device with no company-operated backend.

10.2 MVP vs Full System Comparison

Capability	MVP	Scalable
Apple / Google Sign-In		
Team invitations	—	
On-device OCR		
Batch card capture	—	
Local SwiftData		
iCloud synchronisation	—	
AI follow-up (Gemini)		
Custom AI prompt templates	—	
Static template library	(personal)	(personal + team)
Email via URL scheme		(fallback)
In-app Gmail API send	—	
Fixed status triad		—
Custom pipeline stages	—	
Free-text tags		
CSV export / import		
Local reminders	—	
Personal dashboard		
Team dashboard	—	

10.3 MVP Architecture (Simplified)



10.4 Evolution Plan: MVP → Scalable

Phase	Scope	Effort
Phase A	iCloud sync — annotate SwiftData for CloudKit, add SyncService, migrate local data	Moderate
Phase B	In-app email — Gmail API OAuth, MFMailComposeViewController, keep URL scheme as fallback	Moderate
Phase C	Team workspaces — TeamService, SharedTemplate, PipelineStage, CloudKit shared DB, serverless coord	Substantial

11. Testing

11.1 Testing Strategy

Level	Scope	Tools
Unit Testing	Domain services (AuthService, OCRService, LeadService, AIService, CSVService)	XCTest, mocked dependencies
Integration Testing	ViewModel Service SwiftData	XCTest with in-memory SwiftData
UI Testing	Full user flows	XCUITest
Performance Testing	OCR latency, memory, CPU	Xcode Instruments
Security Testing	Keychain usage, TLS, ATS	Static analysis, SwiftLint, manual review
Compatibility Testing	Device matrix	iPhone SE 3, 13, 15 Pro, 16 Pro Max × iOS 17.0, 17.5, 18.0

11.2 Test Cases

TC-ID	Feature	Test Case	Input	Expected Result	Priority
TC-01	Auth	Apple Sign-In success	Valid Apple ID	User profile created, navigate to dashboard	High
TC-02	Auth	Google Sign-In success	Valid Google account	User profile created, navigate to dashboard	High
TC-03	Auth	Sign-In cancelled	User cancels auth	Remain on login screen, no error	Medium
TC-04	Auth	Logout	Tap logout	Session cleared, return to login, data retained	High
TC-05	OCR	Scan clear card	Well-lit business card photo	All 6 fields extracted correctly	High
TC-06	OCR	Scan blurry card	Poor quality image	Partial fields extracted, manual edit available	High

TC-ID	Feature	Test Case	Input	Expected Result	Priority
TC-07	OCR	Gallery import	Select image from gallery	OCR processes, review screen shown	Medium
TC-08	OCR	Performance	Standard card image	2 seconds extraction time	High
TC-09	Lead	Create lead	Save from review screen	Lead persisted in SwiftData, appears in list	High
TC-10	Lead	Edit lead	Modify fields on profile	Changes persisted, updatedAt updated	Medium
TC-11	Lead	Delete lead	Confirm deletion	Lead removed from store and list	High
TC-12	Lead	Search leads	Type partial name	Matching leads shown, case-insensitive	Medium
TC-13	Lead	Status update	Change to “contacted”	Status badge updated, dashboard reflects change	High
TC-14	AI	Generate follow-up	Lead with all fields + notes	1+ draft variations returned with subject + body	High
TC-15	AI	Gemini unreachable	Network off	Error message shown, static templates offered	High
TC-16	AI	Response latency	Standard prompt	5 seconds on 4G	Medium
TC-17	Email	Send via Gmail	Tap Send, Gmail installed	Gmail opens with pre-filled to/subject/body	High
TC-18	Email	Send via Apple Mail	Tap Send, default client	Mail.app opens with pre-filled fields	High

TC-ID	Feature	Test Case	Input	Expected Result	Priority
TC-19	Email	Copy to clipboard	Tap Copy	Message body on pasteboard, success feedback	Medium
TC-20	Email	Mark Sent	Tap Mark Sent	Lead status → “contacted”	High
TC-21	CSV	Export	Tap Export CSV	CSV file generated, share sheet opens	Medium
TC-22	CSV	Import valid	Select valid CSV	Leads appended, count matches	Medium
TC-23	CSV	Import malformed	Select broken CSV	Error shown, no data corrupted	Medium
TC-24	Dashboard	Lead counts	3 new, 2 follow-up, 1 contacted	Chart shows correct distribution	Medium
TC-25	Security	API key storage	Inspect Keychain	Key stored with correct protection class	High
TC-26	Security	Network traffic	Proxy inspection	All calls over TLS 1.2+, no plaintext	High
TC-27	Perf	Memory usage	Scan 20 cards sequentially	Peak memory 400 MB	Medium
TC-28	All	Accessibility audit	Run Xcode audit	Zero critical violations	Medium

11.3 Requirements Traceability Matrix

Every MVP functional requirement maps to at least one test case, ensuring verifiable coverage. The matrix also enables impact analysis: when a requirement changes, the affected test cases are immediately identifiable.

FR ID	Requirement Summary	Test Case(s)	Priority
FR-1.1	Sign in with Apple	TC-01	High
FR-1.2	Sign in with Google	TC-02	High

FR ID	Requirement Summary	Test Case(s)	Priority
FR-1.3	Persist user profile locally	TC-01, TC-02	High
FR-1.4	Logout control	TC-04	High
FR-2.1	Scan screen (camera + gallery)	TC-05, TC-07	High
FR-2.2	Invoke VNRecognize-TextRequest	TC-05, TC-06, TC-08	High
FR-2.3	Parse recognised text into fields	TC-05, TC-06	High
FR-2.4	Review Contact screen	TC-05, TC-06	High
FR-2.5	Free-form tag attachment	TC-09	Medium
FR-3.1	Persist leads in SwiftData	TC-09	High
FR-3.2	Card-based scrollable list	TC-09	High
FR-3.3	Case-insensitive search	TC-12	Medium
FR-3.4	Open Lead Profile on tap	TC-09	High
FR-3.5	Status editing (new / follow-up / contacted)	TC-13	High
FR-3.6	Lead deletion with confirmation	TC-11	High
FR-4.1	Generate Follow-Up control	TC-14	High
FR-4.2	Dispatch prompt to Gemini	TC-14, TC-16	High
FR-4.3	Present draft variations	TC-14	High
FR-4.4	Static template library	TC-15	High
FR-4.5	~300-word body cap	TC-14, TC-17	Medium
FR-5.1	Copy control	TC-19	Medium
FR-5.2	Pre-filled email URL	TC-17, TC-18	High
FR-5.3	Gmail URL scheme	TC-17	High
FR-5.4	Apple Mail URL scheme	TC-18	High
FR-5.5	Mark Sent → contacted	TC-20	High
FR-6.1	Settings (client selector + logout)	TC-04, TC-17, TC-18	High
FR-6.2	CSV export	TC-21	Medium
FR-6.3	CSV import	TC-22, TC-23	Medium
FR-7.1	Dashboard counts + chart	TC-24	Medium

Non-functional coverage:

NFR ID	Requirement	Test Case
NFR-P1	OCR 2 s	TC-08
NFR-P2	Memory 400 MB	TC-27
NFR-P6	AI round-trip 5 s	TC-16
NFR-S1	AES-256 at-rest	TC-25
NFR-S2	Keychain API key storage	TC-25
NFR-S3	TLS 1.2+ enforced	TC-26
NFR-U3	Accessibility audit	TC-28

12. Deployment Plan

Step	Action	Details
1	Code Freeze	Tag release branch, no new features
2	Final Testing	Run full test suite on device matrix
3	TestFlight (Internal)	Upload via Xcode → App Store Connect → Internal testers
4	TestFlight (External)	Invite 5–10 sales professionals for beta feedback
5	Bug Fix Sprint	Address critical feedback, regression test
6	App Store Submission	Submit build + metadata (screenshots, description, privacy labels)
7	App Review	Apple review (~24–48 hrs). Address rejections if any
8	Release	Approve for sale, phased rollout (optional)
9	Post-Launch Monitoring	Crash reports (Xcode Organizer), user feedback

Privacy Labels Required: - Data linked to user: email, name (from sign-in) - Data not linked to user: none - Data used to track: none

13. Maintenance Plan

Area	Plan
Bug Fixes	Hotfix releases within 72 hours for critical bugs
iOS Updates	Test on each new iOS beta; update SDK usage before GA release
Dependency Updates	Monitor Apple SDK + Google SDK changelogs quarterly
Gemini API	Track API versioning; abstract behind protocol for model swaps
Security Patches	Immediate release for any identified vulnerability
Feature Updates	Scalable phases (A → B → C) per evolution plan in §10.4
Data Migration	SwiftData lightweight migrations; CloudKit schema versioning
User Support	In-app feedback form; App Store review monitoring
Analytics	Opt-in anonymised crash reporting; no third-party SDK without privacy review

14. Documentation and References

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End of Complete SRS — Reach Dog v2.0